

Newton's Backward difference Method

Let $y = f(x)$ be a function which takes the values $y_0, y_1, \dots, y_n$ for the values $x_0, x_1, \dots, x_n$ of the independent variable x .

It is required to find a function $\phi(x)$ such that $f(x)$ and $\phi(x)$ agree at these points. The n th degree polynomial $\phi(x)$ may be written as

$$\begin{aligned}\phi(x) = & a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2) + \dots \\ & + a_n(x - x_0)(x - x_1)\dots(x - x_{n-1}) \dots \dots \dots (i)\end{aligned}$$

Now replacing the value of x by $x_0, x_1, \dots, x_n$ one by one, we get

$$\phi(x_n) = a_0 \Rightarrow a_0 = y_n \quad a_1 = \frac{y_1 - y_0}{h} = \frac{\nabla y_n}{h} \quad a_2 = \frac{\nabla^2 y_n}{2!h^2} \quad a_n = \frac{\nabla^n y_n}{n!h^n}$$

Backward difference method cont

- Substituting these values in (i), we get,

$$\begin{aligned}\phi(x) = & y_n + \frac{\nabla y_n}{h}(x - x_0) + \frac{\nabla^2 y_n}{2!h^2}(x - x_0)(x - x_1) + \frac{\nabla^3 y_n}{3!h^3}(x - x_0)(x - x_1)(x - x_2) + \dots \\ & + \frac{\nabla^n y_n}{n!h^n}(x - x_0)(x - x_1)\dots(x - x_{n-1})\dots\dots\dots(ii)\end{aligned}$$

Now put $x = x_n + ph$ i.e. $x - x_n = ph$ $x - x_{n-1} = (p+1)h, \dots, x - x_1 = (p+n-1)h$
in (ii), we get

Backward difference formula cont

- Backward difference formula of Newton

$$\begin{aligned}\phi(x) = & y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots \\ & + \frac{p(p+1)\dots(p+n-1)}{n!} \nabla^n y_n \dots \dots \dots (iii)\end{aligned}$$

Backward difference formula cont

For the following table of values, estimate $f(7.5)$.

x	1	2	3	4	5	6	7	8
$y = f(x)$	1	8	27	64	125	216	343	512

The sales for the last five years is given in the table below. Estimate the sales for the year 1979

Year	1974	1976	1978	1980	1982
Sales (in lakhs)	40	43	48	52	57

Consider the following table of values

x	1	1.1	1.2	1.3	1.4	1.5
$F(x)$	2	2.1	2.3	2.7	3.5	4.5

Use Newton's Backward Difference Formula to estimate the value of $f(1.45)$.

Backward difference formula cont

For the following table of values, estimate $f(7.5)$.

x	1	2	3	4	5	6	7	8
$y = f(x)$	1	8	27	64	125	216	343	512

x	$y = f(x)$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	1				
2	8	7			
3	27	19	12		
4	64	37	18	6	
5	125	61	24	6	0
6	216	91	30	6	0
7	343	127	36	6	0
8	512	169	42	6	0

Backward difference formula cont

$$y_x = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n$$

lem,

$$p = \frac{x - x_n}{h} = \frac{7.5 - 8.0}{1} = -0.5$$

$$\nabla y_n = 169, \quad \nabla^2 y_n = 42, \quad \nabla^3 y_n = 6$$

$$y_{7.5} = 512 + (-0.5)(169) + \frac{(-0.5)(0.5)}{2}(42)$$

$$+ \frac{(-0.5)(0.5)(1.5)}{6}(6)$$

$$= 512 - 84.5 - 5.25 - 0.375$$

$$= 421.875$$

Lagrange Interpolation

The Interpolation Problem

Given a set of $n+1$ points:

$$(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))$$

Find an n^{th} order polynomial: $f_n(x)$

that passes through all points, such that:

$$f_n(x_i) = f(x_i) \quad \text{for } i = 0, 1, 2, \dots, n$$

Lagrange Interpolation

Problem:

Given

x_i	x_0	x_1		x_n
y_i	y_0	y_1		y_n

Find the polynomial of least order $f_n(x)$ such that:

$$f_n(x_i) = f(x_i) \quad \text{for } i = 0, 1, \dots, n$$

Lagrange Interpolation Formula:

$$f_n(x) = \sum_{i=0}^n f(x_i) \ell_i(x)$$

$$\ell_i(x) = \prod_{j=0, j \neq i}^n \frac{(x - x_j)}{(x_i - x_j)}$$

Lagrange Interpolation

$\ell_i(x)$ are called the cardinals.

The cardinals are n^{th} order polynomials :

$$\ell_i(x_j) = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Lagrange Interpolation Example

$$P_2(x) = f(x_0)\ell_0(x) + f(x_1)\ell_1(x) + f(x_2)\ell_2(x)$$

$$\ell_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-1/4)(x-1)}{(1/3-1/4)(1/3-1)}$$

$$\ell_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1/3)(x-1)}{(1/4-1/3)(1/4-1)}$$

$$\ell_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1/3)(x-1/4)}{(1-1/3)(1-1/4)}$$

$$P_2(x) = 2\{-18(x-1/4)(x-1)\} - 1\{16(x-1/3)(x-1)\} \\ + 7\{2(x-1/3)(x-1/4)\}$$

x	1/3	1/4	1
y	2	-1	7

Interpolating Polynomial Using Lagrange Interpolation Method

$$f_4(x) = \sum_{i=0}^4 f(x_i) \ell_i = \ell_0 + 3\ell_1 + 2\ell_2 + 5\ell_3 + 4\ell_4$$

$$\ell_0 = \frac{(x-1)}{(0-1)} \frac{(x-2)}{(0-2)} \frac{(x-3)}{(0-3)} \frac{(x-4)}{(0-4)} = \frac{(x-1)(x-2)(x-3)(x-4)}{24}$$

$$\ell_1 = \frac{(x-0)}{(1-0)} \frac{(x-2)}{(1-2)} \frac{(x-3)}{(1-3)} \frac{(x-4)}{(1-4)} = \frac{x(x-2)(x-3)(x-4)}{-6}$$

$$\ell_2 = \frac{(x-0)}{(2-0)} \frac{(x-1)}{(2-1)} \frac{(x-3)}{(2-3)} \frac{(x-4)}{(2-4)} = \frac{x(x-1)(x-3)(x-4)}{4}$$

$$\ell_3 = \frac{(x-0)}{(3-0)} \frac{(x-1)}{(3-1)} \frac{(x-2)}{(3-2)} \frac{(x-4)}{(3-4)} = \frac{x(x-1)(x-2)(x-4)}{-6}$$

$$\ell_4 = \frac{(x-0)}{(4-0)} \frac{(x-1)}{(4-1)} \frac{(x-2)}{(4-2)} \frac{(x-3)}{(4-3)} = \frac{x(x-1)(x-2)(x-3)}{24}$$

Divided Differences

$$f[x_k] = f(x_k)$$

Zeroth order DD

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

First order DD

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

Second order DD

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$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

Divided Difference Table

x	F[]	F[,]	F[, ,]	F[, , ,]
x ₀	F[x ₀]	F[x ₀ , x ₁]	F[x ₀ , x ₁ , x ₂]	F[x ₀ , x ₁ , x ₂ , x ₃]
x ₁	F[x ₁]	F[x ₁ , x ₂]	F[x ₁ , x ₂ , x ₃]	
x ₂	F[x ₂]	F[x ₂ , x ₃]		
x ₃	F[x ₃]			

$$f_n(x) = \sum_{i=0}^n \left\{ F[x_0, x_1, \dots, x_i] \prod_{j=0}^{i-1} (x - x_j) \right\}$$

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

x_i	$f(x_i)$
0	-5
1	-3
-1	-15

Entries of the divided difference table are obtained from the data table using simple operations.

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

x_i	$f(x_i)$
0	-5
1	-3
-1	-15

The first two column of the table are the data columns.

Third column: First order differences.

Fourth column: Second order differences.

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

x_i	
0	-5
1	-3
-1	-15

$$\frac{-3 - (-5)}{1 - 0} = 2$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

$$\frac{-15 - (-3)}{-1 - 1} = 6$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$$

x_i	
0	-5
1	-3
-1	-15

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

x_i	
0	-5
1	-3
-1	-15

$$\frac{6 - (2)}{-1 - (0)} = -4$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

Divided Difference Table

x	F[]	F[,]	F[, ,]
0	-5	2	-4
1	-3	6	
-1	-15		

x_i	
0	-5
1	-3
-1	-15

$$f_2(x) = -5 + 2(x-0) - 4(x-0)(x-1)$$

$$f_2(x) = F[x_0] + F[x_0, x_1](x-x_0) + F[x_0, x_1, x_2](x-x_0)(x-x_1)$$

Example

- The upward velocity of a rocket is given as a function of time in the following table

$t \text{ (s)}$	$v(t) \text{ (m/s)}$
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

- a) Determine the value of the velocity at $t = 16 \text{ seconds}$ with *third order polynomial* interpolation using Newton's divided difference polynomial method.
- b) Using the third order polynomial interpolant for velocity, find the distance covered by the rocket from $t = 11 \text{ s}$ to $t = 16 \text{ s}$.
- c) Using the third order polynomial interpolant for velocity, find the acceleration of the rocket at $t = 16 \text{ s}$.

Example

x	f(x)	f[,]	f[, ,]	f[, , ,]
10	227.04			
15	362.78			
20	517.35			
22.5	602.97			

Example

x	f(x)	f[,]	f[, ,]	f[, , ,]
10	227.04	27.148	0.37660	5.4347×10^{-3}
15	362.78	30.914	0.44453	
20	517.35	34.248		
22.5	602.97			

Example

$$\begin{aligned}v(t) &= b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) + b_3(t - t_0)(t - t_1)(t - t_2) \\&= 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15) \\&\quad + 5.5347 \times 10^{-3}(t - 10)(t - 15)(t - 20) \\&= -4.2541 + 21.265t + 0.13204t^2 + 0.0054347t^3, \quad 10 \leq t \leq 22.5\end{aligned}$$

At $t = 16$,

$$\begin{aligned}v(16) &= 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) \\&\quad + 5.5347 \times 10^{-3}(16 - 10)(16 - 15)(16 - 20) \\&= 392.06 \text{ m/s}\end{aligned}$$

The distance covered by the rocket between $t = 11 \text{ s}$ and $t = 16 \text{ s}$ can be calculated from the interpolating polynomial

$$s(16) - s(11) = \int_{11}^{16} v(t) dt = 1605 \text{ m}$$